

Feature Documentation

Financial Features (Primary Drivers)

These features carry the **strongest influence** on valuation prediction.

Feature	Data Type	Meaning	Example	Value Range	Validation Rule	Why It Matters	Model Contribution
annual_revenue_egp	float	Total yearly revenue	6,364,369	100,000 → 100,000,000	Must be > 0	Represents business size	Higher revenue increases valuation
annual_expenses_egp	float	Total yearly expenses	5,538,994	50,000 → 90,000,000	Must be ≤ revenue	Determines profitability	Higher expenses reduce net profit
annual_profit_egp	float	Net profit after expenses	825,374	-5,000,000 → 50,000,000	Can be negative	Indicates financial health	Higher profit increases value
profit_margin	float	Profit divided by revenue	0.1297	-0.5 → 0.6	Must be between -50% and 60%	Measures operational efficiency	Higher margin improves valuation reliability

Business Size & Stability Features

These reflect operational maturity and reliability.

Feature	Data Type	Meaning	Example	Value Range	Validation Rule	Why It Matters	Model Contribution
number_of_employees	int	Number of workers	24	1 → 200	Must be integer ≥ 1	Indicates scale of operations	Larger workforce signals structured business
years_in_operation	float	Years business has operated	8	0 → 30	Can be ≥ 0	Indicates stability	Older businesses seen as safer

Growth Indicators

These estimate **future potential**.

Feature	Data Type	Meaning	Example	Value Range	Validation Rule	Why It Matters	Model Contribution
revenue_growth_rate	float	Revenue change year-over-year	-0.0045	-0.5 → 1.0	Must be between -50% and +100%	Indicates trend direction	Positive growth increases value
customer_growth_rate	float	Customer base	-0.0048	-0.5 → 1.0	Same as revenue	Reflects market	Supports revenue

Feature	Data Type	Meaning	Example	Value Range	Validation Rule	Why It Matters	Model Contribution
		growth rate			growth	demand growth	predictions
monthly_customers	int	Average customers per month	686	10 → 20,000	Must be > 0	Indicates demand strength	Higher demand supports growth

Market & Competition Features

These measure external conditions.

Feature	Data Type	Meaning	Example	Value Range	Validation Rule	Why It Matters	Model Contribution
market_demand_score	int	Demand level in market	10	1 → 10	Must be integer 1–10	High demand supports growth	Higher demand increases valuation
competition_level	int	Competition intensity	7	1 → 10	Must be integer 1–10	Competition reduces profits	Higher competition lowers valuation

Risk & Operational Features

These reflect operational dependency risk.

Feature	Data Type	Meaning	Example	Value Range	Validation Rule	Why It Matters	Model Contributio
owner_dependency_score	int	Dependency on owner	10	1 → 10	Must be integer 1–10	High dependency increases risk	Higher dependency lowers valuation
revenue_stability_score	int	Revenue consistency	8	1 → 10	Must be integer 1–10	Stability improves reliability	Higher stability increases trust

Scalability & Location Features

These measure future expansion capability.

Feature	Data Type	Meaning	Example	Value Range	Validation Rule	Why It Matters	Model Contribution
scalability_score	int	Ability to scale operations	5	1 → 10	Must be integer 1–10	Scalable businesses grow faster	Higher scalability increases value
location_score	int	Quality of physical location	6	1 → 10	Must be integer 1–10	Location affects customer traffic	Better location improves growth

Categorical Features

Converted into numerical form using **One-Hot Encoding**.

Feature	Data Type	Meaning	Example	Value Range	Validation Rule	Why It Matters	Model Contribution
industry	category	Business type	Service Business	Predefined categories	Must exist in list	Different industries have different valuation patterns	Defines base behavior patterns
city	category	Business location	Cairo	Predefined cities	Must exist in supported list	Cities differ economically	Affects demand & competition

Growth Rate Interpretation Reference

Growth values are stored as **decimal numbers**, not percentages.

Stored Value	Actual Meaning
0.10	+10% growth
0.05	+5% growth
0.00	No growth
-0.05	-5% decline

Example:

revenue_growth_rate = -0.0045

Means:

-0.45% revenue decline

This is **normal behavior** in real businesses.

Feature Relationship Summary (Model Thinking Logic)

The model builds valuation logic using combinations like:

Positive pattern:

High revenue

- Strong profit margin
- Good scalability
- Stable revenue
- = Higher valuation

Negative pattern:

Low growth

- High competition
- High owner dependency
- = Lower valuation

Random Forest builds **hundreds of these decision paths internally**.